MILITARY SPECIFICATION

TIMING MODULE - 11737703

1. SCOPE

1.1 Scope this specification covers one type of electronic assemble known as Timing Module, Part Number (P/N) 11737703.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing The Manufacture And Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army, Materiel: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance, and Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical Electronic, Instrument, Communication, And Radar For Aerospace And Control Systems, Procedure For
MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type
MIL-P-46843	Printed Circuit Assemblies, Design And Production of
MIL-P-55640	Printed Wiring Boards, Multi-Layered (Plated-through Hole)

STANDARDS

Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-810	Environmental Test Methods

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DRAWINGS

U.S. Army, Frankford Arsenal

F11737703	Timing Module

PACKAGING DATA SHEET

11737703	Packaging of Timing Module 11737703

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication - This module shall be manufactured in accordance with Drawing F11737703 and all drawings pertaining thereto.

3.2 Feneral specification. - The timing module shall meet the following requirements, where applicable, of MIL-F-13926:

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Electrical and electronic assemblies
e. Part identification for marking
f. Workmanship

3.2.1 Printed circuit assemblies. - Printed circuit assemblies shall be in accordance with MIL-P-46843 and MIL-P-55640.

3.2.2 Flow soldering. - Flow soldering shall be in accordance with MIL-S-46844 and MIL-S-45743.

3.3 PERFORMANCE

3.3.1 Interchangeability. - The timing module shall meet all of the requirements of this specification and shall be interchangeable with all other timing modules such that the replacement of any module with any other module shall in no way cause incapability wiht the requirements of this specification.

3.3.2 Environmental.

3.3.2.1 Temperature storagw. - Module to meet requirements of 3.3.3 subsequent to being exposed and thermally stabilized at temperatures of -65° and +155° Fahrenheit (F).

3.3.2.2 Operating temperature. - Module to meet requirements of 3.3.3 while being exposed to temperatures of 0°F to 100°F.

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3.3.2.3 Vibration. - Module to be mounted as used. Test per MIL-STD-810, Method 514.2, Procedure VIII, Curve V, Figure 514.2-6, resonance dwell not required. Module shall meet the requirements of 3.3.3 subsequent to the test.

3.3.2.4 Shock. - Test per MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-1, Note 1 Module shall meet requirements of 3.3.3 subsequent to this test.

3.3.25. Humidity. - Test per MIL-STD-810, Method 507.1, Produce IV, Figure 507.1-3. Chamber temperature not to exceed 155°F. Module to meet requirements of 3.3.3 subsequent to this test.

3.3.2.6 Fungus. - Test per MIL-STD-810, Method 508.1, Procedure I. Module to meet requirement of 3.3.3 subsequent to this test.

3.3.3 Performance-operation. - When the following voltages are applied simultaneously, the output waveform at TP-3 shall conform to Figure 1. Voltages: 24 ± 6 VDC applied to Pin "M" (TP-1), 135 ± 10 VDC applied to Pin "K" and 14 ± 1 VDC applied to Pin "A" through a 1K, 1/4 watt resistor.

OUTPUT WAVEFORM TP-3

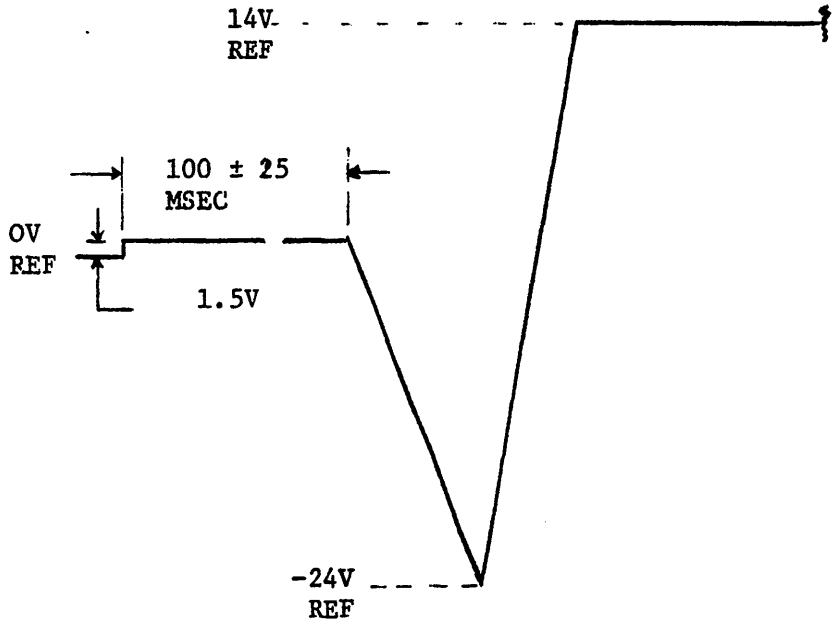


FIGURE 1

I

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4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed. necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First Article (initial production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for first reticle testing shall be as specified in the contract. The sample for first article approval tests shall consist of three assemblies manufactured in the shine manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production. The three first article article samples will be submitted to all the tests indicated in Table I and Table II. Of the tests indicated in Table III and Table IV, one sample shall be subjected to the Fungus Test (4.6.2.7) and the other samples shall be subjected to the other test specified.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

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4.3.2.1 Components and subassemblies. - All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP'S) listed in the Technical Data Package (TDP). In the absence of SQAP'S, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein, shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

(100% INSPECTION)

No.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Interchangeability	3.3.1	4.6.1
102.	Performance - operation	3.3.3	4.6.3

NOTE: The tests in Table I shall be conducted at a temperature between +70°F and +80°F.

4.3.3 Acceptance and rejection. - Rejection lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification. - Certification shall include test data and results for the characteristics specified below. Certificaiton shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract productions including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

NO.	CHARACTERISTIC	TEST DATA TO COMPLY WITH
301.	Cleaning	MIL-P-46843
302.	Drying	"
303.	Coating, insulating compound	"

4.4 Special sampling.

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4.4.1 General.- 4.4.1 General. - Three assemblies shall be selected at random as a special sampling from each 50 produced. The sample shall meet the re- The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401.	Materials	3.1	Applicable dwgs. & specs - visual
402.	Fabrication	3.1	Applicable dwgs. & specs - visual
403.	General specification	3.2	MIL-F-13926 - visual, MIL-P-46843

NOTE: The tests in Table II shall be conducted at a temperature between +70°F and +80°F.

4.4.2 Environmental. - Three assemblies shall be selected at random as samples from each 50 produced, for from each month's production, whichever occure first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
404.	Storage temperature (-65°F to 155°F)	3.3.2.1	4.6.2.1
405.	Operating temperature (0°F to 100°F)	3.3.2.2	4.6.2.2
406.	Vibration	3.3.2.3	4.6.2.3
407.	Shock	3.3.2.4	4.6.2.4

4.4.2.1 Control tests. - Three assemblies shall be selected and meet the requirements and tests of Table IV. These tests shall be conducted as follow-up tests to the fires article samples at 25%, 50% and 75% points of completionof the production contract.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
501.	Humidity	3.3.2.5	4.6.2.5
502.	Fungus	3.3.2.6	4.6.2.6

4.4.3 Failure of samPle. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

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4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - When the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment. -

4.5.2.1 Government design. - All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this. inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and Procedures.

4.6.1 Interchangeability. - This test shall be performed utilizing an electronic enclosure assembly, Drawing 11737724, to meet the requirements of 3.3.1.

4.6.2 Environmental. - The assemblies shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the assemblies shall be subjected to the tests specified in 4.6.1 and 4.6.3.

4.6.2.1 Storage temperatures. - The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.3.2.1.

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4.6.2.2 Operational temperatures. - The test shall be performed in accordance with Procedure I of MIL-F-13926 at the extreme temperatures specified in 3.3.2.2 utilizing the testing equipment indicated in 4.6.2.1. The assembly shall be exposed to each specified temperature for a period of twe hours. At the end of each two hour period and while still at the specified temperature, the assembly shall be subjected to the tests in 4.6.3. The assembly shall conform to the requirements of 3.3.2.2, and 3.3.3.

4.6.2.3 Vibration. - This test shall be performed as per MIL-STD-810, Method 514.2, Procedure VIII, Curve V, Figure 514.2-6, resonance dwell not required. Subsequent to the vibration test, the module shall meet the requirements of 3.3.3.

4.6.2.4 Shock. - This test shall be performed as per MIL-STD-810, Method 516.2, Procedure I, Figure 516.2-1, Note 1. Subsequent to the shock test, the module shall meet the requirements of 3.3.3.

4.6.2.5 Humidity. - This test shall be performed as per MIL-STD-810, Method 507.1, Procedure IV, Figure 507.1-3. Subsequent to the test the module shall meet the requirements of 3.3.3.

4.6.2.6 Fungus. - This test shall be performed as per MIL-STD-810, Method 508.1, Procedure I. Subsequent to the fungus exposure the module shall meet the requirements of 3.3.3.

4.6.3 Performance operation. - Perform this test utilizing standard test equipment to supply the voltages specified in 3.3.3 and measure the waveform as shown in Figure I.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11737703. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Timing Module is a functional assembly of the LASER Trainer: M55.

6.2 Ordering data. - Procurement documents should specify the foilowing:

(a) Title, number, and date of this specification
(b) Application packaging data sheet number (see 5.1)
(c) Selection of applicable levels of preservation, packaging and packing
(d) Applicable stock number
(e) Requirement for First Article Submission

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Preparing Activity:

Army - MU

project No. 1240-A674

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-T-48409A(MU) Timing Module - 11737703

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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